

Assessing the forecast skill of a stochastic weather prediction model for extreme precipitation events in India

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1 Background and objective of the project

An extreme precipitation event is defined statistically as an occurrence where the rainfall at a particular place and time exceeds some predefined threshold (usually the 95th or 99th percentile) of a long-term climatological record for that location. These are a major environmental threat, as they often lead to flooding, landslides, and cause severe socio-economic disruption in the affected regions. Thus an accurate and reliable prediction of such events is essential for devising effective intervention measures.

1.1 Deterministic v/s Probabilistic forecasting

The atmosphere – being an inherently chaotic system – is very sensitive to even small uncertainties in its initial state, which can amplify quickly over time and lead to vastly different trajectories. Given that deterministic forecasts are, by design, constrained to a single evolution path, they may completely miss out on a small instability that later causes or develops into an extreme event. Thus probabilistic or ensemble forecasting is preferred in such cases. They generate a range of possible future scenarios as forecast output, using which one can quantify the likelihood of different outcomes and the “spread”; this provides us with a more complete picture of the future atmospheric state compared to a deterministic forecast.

1.2 Methods of ensemble generation

Traditionally, such ensembles have been generated using deterministic NWP models which assume that forecast uncertainty arises mainly from an imperfect knowledge about the atmosphere’s initial state. To do this, a single *best-estimate* analysis of the atmospheric conditions is first constructed. Then systematic perturbations are introduced to create multiple plausible starting conditions. These perturbations are carefully designed using techniques like singular vectors or ensemble of data assimilations (EDA), to target areas where error grows the fastest. The full NWP model is then run separately for each starting point to create an ensemble of future predictions. While this method does provide physically robust forecasts, it has a high computational cost and often requires hours even on supercomputers to generate even a medium-sized ensemble.

Unlike traditional NWP ensembles, a new generation of machine-learning-based models using neural networks can generate ensemble forecasts much more efficiently. These models inject a stochastic noise vector into one of the network’s internal layers at each forecast step; during training, the model learns to shape these noise vectors into physically consistent forecast trajectories. Ensemble generation using this method can often be accomplished in a few minutes even on a small number of GPUs.

1.3 Objective of this project

Although this method of ensemble generation is very computationally efficient, it is still important to evaluate the skill and reliability, and see whether the forecasts (especially for the extreme events) are accurate, physically plausible, and produce a realistic ensemble spread.

Previously, AI-based models have not shown great skill in predicting extreme rainfall, mostly due to the prevalence of a lot of zero-rainfall samples in the training data set. To assess whether some modern stochastic approaches can better represent these extremes, here we evaluate the performance of the machine-learning-based ensemble prediction system *aifs-ens-1.0* for ten documented extreme rainfall events across diverse regions of India. For each case, a 10-member ensemble was initialized five days prior to the event date. The analysis then evaluates the forecast against the corresponding ERA5 reanalysis precipitation fields for each event.

2 About aifs-ens-1.0 (AIFS-CRPS)

The model used in this analysis is *aifs-ens-1.0* (also referred to as AIFS-CRPS). This is a machine-learning-based model developed at the European Centre for Medium-Range Weather Forecasts (ECMWF) specifically for probabilistic weather prediction.

2.1 Training objective and loss functions

A main distinction between machine-learning-based weather models is the objective function that they are trained on. Many early models were trained to minimize a Mean-Squared-Error (MSE) loss. An issue was that this approach incentivises the model to produce spatially smooth forecasts that average out fine-scale details and thus under-predict the magnitude of extreme event (the *double penalty problem*). AIFS-CRPS is instead trained on a refined version of the Continuous Ranked Probability Score (CRPS):

1. **The Continuous Ranked Probability Score (CRPS):** Given an ensemble forecast $\{x_j\}_{j=1}^M$ and a verifying observation y , the CRPS is defined as follows:

$$CRPS(\{x_j\}, y) = \frac{1}{M} \sum_{j=1}^M |x_j - y| - \frac{1}{2M^2} \sum_{j=1}^M \sum_{k=1}^M |x_j - x_k| \quad (1)$$

The first term $\frac{1}{M} \sum_{j=1}^M |x_j - y|$ represents the average absolute error of the ensemble members relative to the observation. The second term $\frac{1}{2M^2} \sum_{j=1}^M \sum_{k=1}^M |x_j - x_k|$ is a measure of the ensemble spread by averaging the pairwise distances between all members. Together these components ensure that the CRPS rewards forecasts that are both accurate and properly calibrated (lower values indicating better performance)

2. **The Fair CRPS (fCRPS):** The standard CRPS is known to be statistically biased for finite ensemble sizes. The fCRPS corrects for this bias:

$$fCRPS(\{x_j\}, y) = \frac{1}{M} \sum_{j=1}^M |x_j - y| - \frac{1}{2M(M-1)} \sum_{j=1, k \neq j}^M |x_j - x_k| \quad (2)$$

However, the fCRPS suffers from a degeneracy issue – if all but one member perfectly match the truth, the loss for the final member will become zero irrespective of its value.

3. **The Almost Fair CRPS (afCRPS):** To get the best of both worlds – most of the fairness of fCRPS and stability of CRPS – the model is trained on a weighted average of the two scores (the afCRPS):

$$afCRPS := \alpha \cdot fCRPS + (1 - \alpha) \cdot CRPS \quad (3)$$

For this particular model, $\alpha = 0.95$.

2.2 Model architecture and training data

The AIFS-CRPS architecture follows an encoder-processor-decoder design. Both the encoder and decoder are transformer-based graph neural networks (GNNs), and the processor is a sliding window transformer. The model was trained on the Copernicus ERA5 reanalysis dataset using global atmospheric data from 1979-2017.

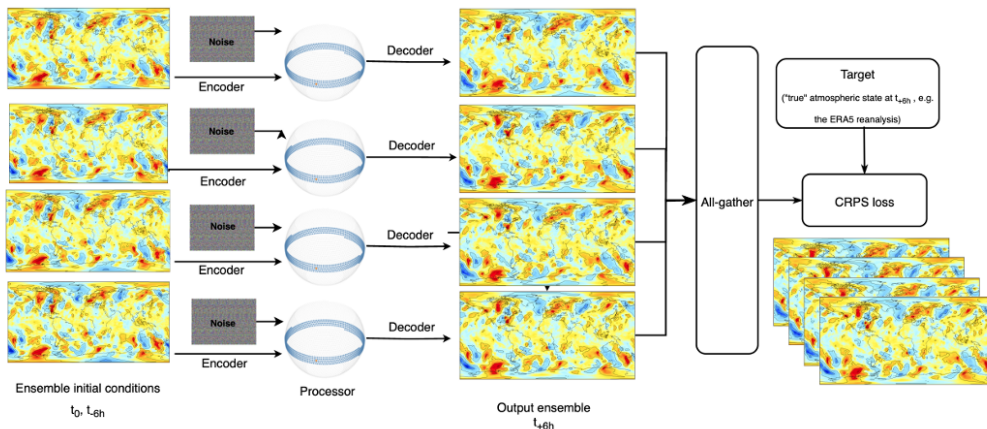


Figure 1: Probabilistic training of AIFS-CRPS

2.3 Input and output variables

The table below provides a list of all input and output variables (alongwith their atmospheric levels), and whether they are used as inputs, outputs, or both; and the normalisation applied to each field:

Field / Variable	Level type	Input/Output	Normalisation
Geopotential, horizontal and vertical wind components, specific humidity, temperature	Pressure level: 50, 100, 150, 200, 250, 300, 400, 500, 600, 700, 850, 925, 1000	Both	Standardised, apart from the geopotential, which is max-scaled
Surface pressure, mean sea-level pressure, skin temperature, 2 m temperature, 2 m dewpoint temperature, 10 m horizontal wind components, total column water	Surface	Both	Standardised
Total precipitation	Surface	Output	Standard deviation changed but mean kept the same
Land-sea mask, orography, standard deviation and slope of sub-grid orography, insolation, cosine and sine of latitude and longitude, cosine and sine of the local time of day and day of year	Surface	Input	Sub-grid orography max-scaled; all others not normalised
Time step identifier	-	Input	-

Figure 2: Input and output variables of AIFS-CRPS

3 Data and Methodology

3.1 Data

Ten historical cases of extreme precipitation across diverse geographical regions of India were selected for this study. These events were sourced from the XWeatherLib repository [2]. The selected events are given in Table 1. The AIFS-CRPS (aifs-ens-1.0) model was used to generate all forecasts. For both model initialization and forecast verification, the ERA5 reanalysis dataset ECMWF was used as the *ground truth*.

Serial No.	Region	Date of Event
1	Maharashtra	2002-08-25
2	Guwahati	2004-04-03
3	Odisha	2007-08-06
4	Uttarakhand	2013-07-14
5	Kashmir	2014-09-04
6	Chennai	2015-12-01
7	Yamunanagar	2016-07-15
8	Kerala	2018-08-14
9	Kerala	2019-08-08
10	Telangana	2023-03-18

Table 1: List of historical extreme precipitation events selected for analysis.

3.2 Methodology

3.2.1 Protocol

For each event, the following procedure was used:

1. The AIFS-CRPS model was initialized with the ERA5 atmospheric state from **five days prior** to the target date of the event
2. A **10-member ensemble forecast** was generated
3. The 24-hour accumulated precipitation field for the target date was calculated for each of the ten ensemble members.
4. The time evolution of some key precursor variables to extreme rainfall (in particular q_{850} : specific humidity at 850 hPa, w_{500} : vertical velocity at 500 hPa, t_{700} : temperature at 700 hPa, and tcw : total column water) was plotted and compared with the original values.

3.2.2 Evaluation

A strict grid-by-grid comparison would penalize forecasts that correctly predict the event’s magnitude but place it slightly offset, relative to forecasts that predict a sparse rainfall everywhere. Thus instead of evaluating forecasts at the exact grid points where extreme rainfall occurred, we calculate the performance over some region surrounding the event. At the same time, we should also penalise false alarms – because otherwise, a model that predicts extreme rainfall everywhere and on most times of the year would get a good score, although it is obviously unreliable (the *forecaster’s dilemma*).

Inside the chosen evaluation region, all grid points exceeding the 90th percentile of 24-hour accumulated rainfall are classified as *affected*. Each forecast grid point is then labelled as a Hit (H), Miss (M), False Alarm (F), or Correct Negative (CN) by comparing with the ground truth (with the variables having usual meaning). From these counts, we calculate a score or index for each event:

$$P = \frac{H}{H + M + F}. \quad (4)$$

Each member is categorized based on its score as follows (this is a little arbitrary):

- **Skillful:** $P > 0.3$
- **Some Skill:** $0.1 \leq P \leq 0.3$
- **Failed:** $P < 0.1$

4 Results and Analysis

The forecast performance using the above evaluation metric is summarized in the table below:

Event Location	Date	Skillful Members (out of 10)	Members with Some Skill (out of 10)	Failed Members (out of 10)	Ensemble Consensus & Confidence
Yamunanagar	2016-07-15	10	0	0	Extremely High Confidence
Chennai	2015-12-01	10	0	0	Extremely High Confidence
Maharashtra	2002-08-25	7	3	0	High Confidence
Kerala	2019-08-08	10	0	0	High Confidence
Kashmir	2014-09-04	1	1	8	Very Low Confidence
Guwahati	2004-04-03	1	0	9	Very Low Confidence
Telangana	2023-03-18	1	2	7	Low Confidence
Uttarakhand	2013-07-14	0	7	3	Low Confidence
Kerala	2018-08-14	0	0	10	Complete Forecast Failure
Odisha	2007-08-06	0	0	10	Complete Forecast Failure

Figure 3: Summary of the 10-member ensemble’s forecast skill across the ten selected extreme rainfall events

4.1 Precipitation forecast plots

For illustration, three events – Telangana 2023, Kerala 2019, and Odisha 2007 – have been considered, and the first few members plotted for each:

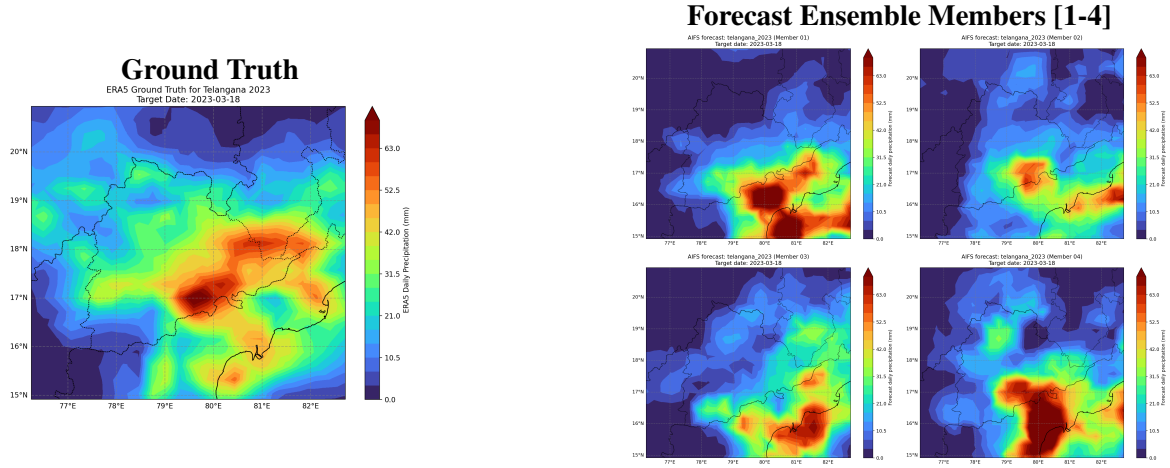


Figure 4: Comparison of ground truth (L) and ensemble forecasts (1-4) (R) for the Telangana 2023 event

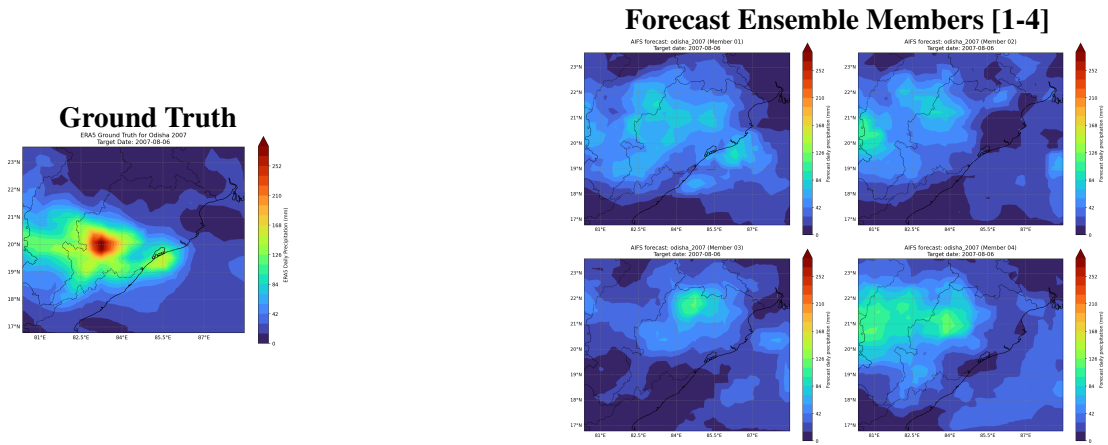


Figure 5: Comparison of ground truth (L) and ensemble forecasts (1-4) (R) for the Odisha 2007 event

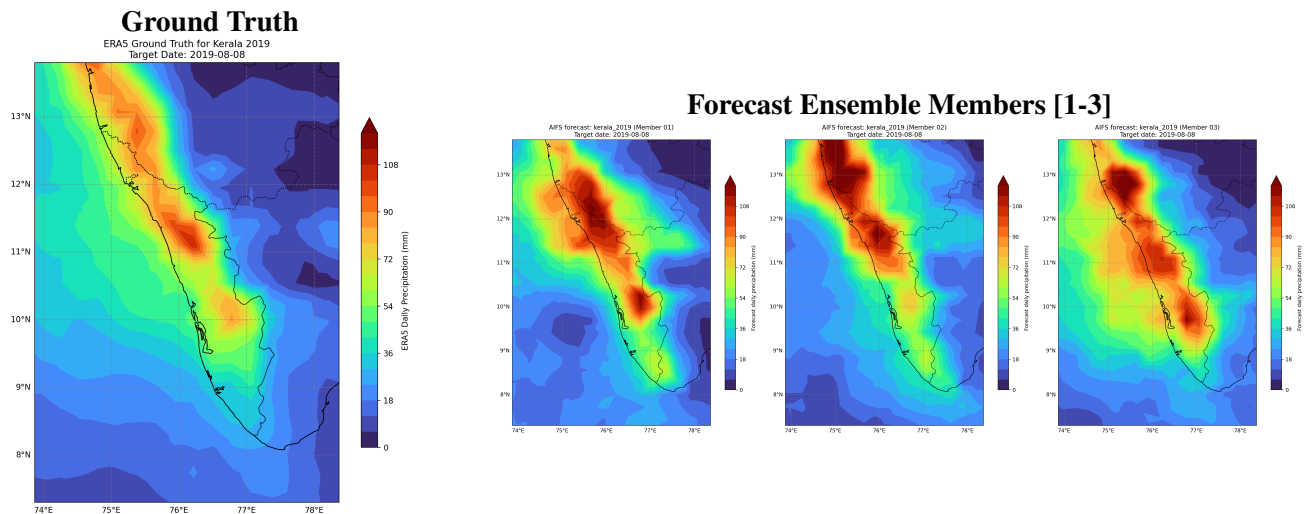


Figure 6: Comparison of ground truth (L) and ensemble forecasts (1-3) (R) for the Kerala 2019 event

4.2 Precursor variables to extreme rainfall

High tcw and $q850$ values show that the atmosphere and lower troposphere hold a lot of moisture. Strong $w500$ (by convention, a negative sign means upward direction) implies upward motion that lifts this moist air and encourages condensation; $t700$ typically decreases as we approach the precipitation event (for simplification, we have considered only one pressure level for each)

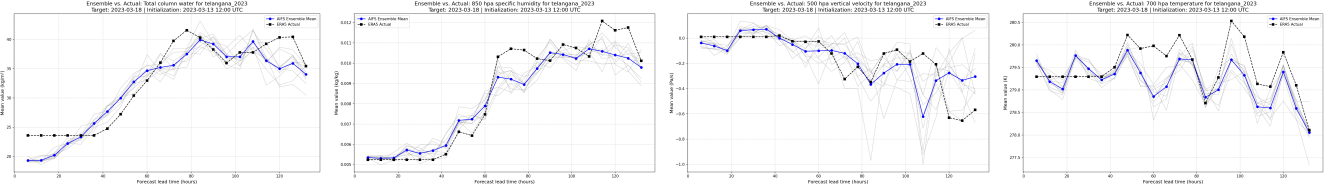


Figure 7: Time series of precursor variables for the Telangana 2023 event. L $\rightarrow$ R: tcw , $q850$, $w500$, and $t700$.

4.3 Analysis of forecast performance

The general pattern in forecast quality could be summarized as follows: events that are driven by large-scale, long-term weather patterns are generally easier to predict; short-lived, localized events, and those particularly occurring outside usual seasonal periods or confined to limited regions are harder to predict.

Particularly for the three cases that we had considered, in the 2023 Telangana event, several members correctly located the main area of precipitation, but they consistently slightly underestimated the peak of rainfall. Whereas, the forecast for the 2019 Kerala event also performed well in terms of locating the event, but many members over-forecasted the magnitude of rainfall.

The 2007 Odisha event was perhaps the largest forecast failure among the cases studied. None of the ensemble members could capture the magnitude of extreme rainfall. Looking at the key precursor variables, the model underestimated the large influx of low-level moisture before the event and also missed the strength of the dynamic trigger. (see Figure 8). It may be speculated that the low forecast skill may have resulted from processes occurring at scales smaller than the model's resolution, so that certain localized moisture surges or mesoscale convective systems were likely underrepresented.

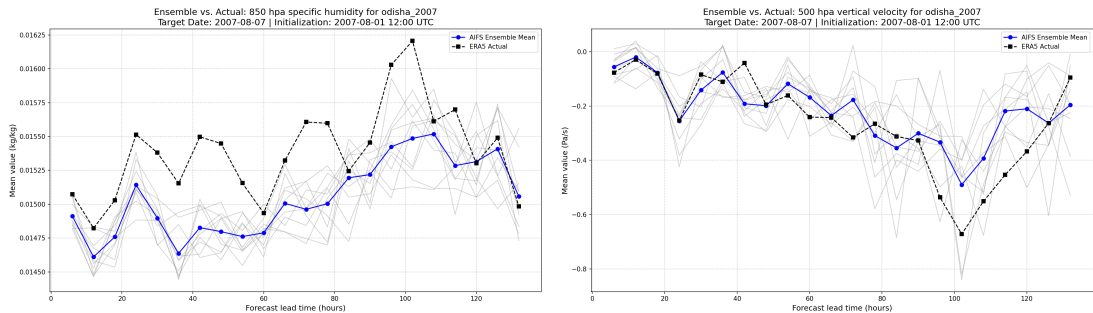


Figure 8: Time series for the Odisha 2007 event showing ensemble forecasts and observations for 850 hPa specific humidity (left) and 500 hPa vertical velocity (right)

References

- [1] Lang, S., Alexe, M., Clare, M. C. A., Roberts, C., Adewoyin, R., Ben Bouall'egue, Z., Chantry, M., Dramsch, J., Dueben, P. D., Hahner, S., Maciel, P., Prieto-Nemesio, A., O'Brien, C., Pinault, F., Polster, J., Raoult, B., Tietsche, S., Leutbecher, M. (2024). AIFS-CRPS: Ensemble forecasting using a model trained with a loss function based on the Continuous Ranked Probability Score (preprint). arXiv.
- [2] XWeatherlib repository, Centre for Climate Studies, Indian Institute of Technology, Bombay